

Bimolecular QRRK Analysis of Methyl Radical Reactions

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Abstract

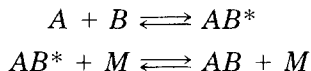
Reactions which proceed through energized adducts, including radical recombinations, insertions, and addition to unsaturates, frequently exhibit unusual kinetic behavior. The branching ratios among various product channels are often complex functions of both temperature and pressure. Four such reactions involving methyl radicals are analyzed by combining chemical activation distribution functions with QRRK methods to predict rate constants for each channel. These include three oxidation paths, $\text{CH}_3 + \text{O}$, $\text{CH}_3 + \text{O}_2$, $\text{CH}_3 + \text{OH}$, and the addition reaction $\text{CH}_3 + \text{C}_2\text{H}_2$. These predictions are compared to experiments wherever possible; generally, the agreement is quite satisfactory. Analysis of the energetics of the various reaction channels, using parameters which are readily available, provides a convenient framework for prediction. Suggested rate constants for the various channels for the four reactions are given at three pressures, 20, 760, and 7600 Torr, for the temperature range 300–2500 K. The approach used here can easily be applied to other reactions.

Introduction

Kineticists are becoming increasingly aware of the necessity of properly treating the pressure dependence of simple dissociation and recombination reactions. Although many of these reactions involving polyatomic molecules are at the high-pressure limit, i.e., pressure-independent, at one atmosphere and low temperatures, this limit shifts to much higher pressures as the temperature is increased toward that normally encountered in combustion. Explicit account must be taken of this fall-off behavior when constructing kinetic models for high-temperature systems, as discussed in the recent review of Warnatz [1]. The shift in the fall-off curves with temperature is simply a manifestation of the fact that these reactions are not simple one-step processes.

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Consider the simple recombination reaction:



The shift is due to the markedly different temperature dependency of dissociation and stabilization. The dissociation rate for the energized complex increases rapidly with temperature so that the number of deactivating collisions must be substantially increased to maintain the high-pressure limiting behavior. One would expect to see similar behavior for any reaction which proceeds via a chemically activated energized complex. Indeed, a variety of reaction types, including additions, recombinations, and insertions, have been analyzed in this way using RRKM theory [2-5].

Another property of reactions proceeding via energized complexes is the possibility of additional dissociation channels to new products. A particularly interesting example is the $\text{CH}_3 + \text{CH}_3$ reaction. It was shown [3] that the products can shift from C_2H_6 (chain termination) to $\text{C}_2\text{H}_5 + \text{H}$ (chain propagation) as the temperature is increased and the complex has enough energy to access the endothermic channel leading to $\text{C}_2\text{H}_5 + \text{H}$. As a result, radical addition, recombination, and insertion reactions can manifest very complex temperature and pressure behavior as the stabilization channel competes with these multiple decomposition channels. This can result in substantial errors in extrapolation of rate constants for these types of reactions to other regimes of temperature and pressure.

Recently, a simpler approach based on QRRK theory and a chemical-activation formalism was devised which permitted more straightforward evaluation of the temperature and pressure behavior of such systems [6]. As with the more sophisticated RRKM treatments, it was shown that endothermic channels could become accessible. Treatment of the addition reaction of methyl radicals to unsaturates showed that the high-temperature kinetics of these systems could be significantly different than commonly envisioned because the complex has enough energy to split out H atoms before collisional stabilization. This path dramatically influences the kinetics of hydrocarbon systems by providing a facile route for conversion of relatively unreactive methyl radicals to highly reactive hydrogen atoms. This QRRK approach has also been used to analyze a variety of reactions where there were ample literature data to properly evaluate the predictions of the method [7]. It was shown to be quite successful in predicting rate constants; in addition, it presented a simple framework by which the effects of temperature and pressure could be readily understood.

The QRRK method is a potent supplement to a RRKM analysis despite being less rigorous. It will be shown that the information required for the QRRK approach is readily available or straightforward

to estimate. In particular, the high-pressure "A-factor" for dissociation of the energized adduct is based on measured values or assigned by comparison to known rate constants. We avoid using RRKM theory to predict this A-factor from the frequency distribution and moments of inertia of the transition state. For many of the systems of interest, the structure of the transition state is sufficiently vague that adjustments of these parameters are required to obtain "reasonable" values, and thus the predictive power is minimal. Thus, although an RRKM analysis is more accurate in principle, the uncertainties in the input data often restrict its predictive powers.

A more serious shortcoming of the QRRK approach is use of a single geometric-mean frequency to represent the frequency distribution. One might expect problems with such an approximation, particularly at lower temperatures. Weston [8] has made extensive comparisons of various formulations of the Kassel approach to RRKM results for a series of thermal unimolecular dissociations. (A point of potential confusion here is that the approach we have chosen to designate QRRK is specified to be QKG by Weston). Weston finds that use of a single geometric-mean frequency tends to overestimate $k(E)$ for energies within 10 Kcal mol⁻¹ of the critical energy for dissociation; the overestimate is worse when the arithmetic-mean frequency is used. Comparison of predictions of fall-off behavior showed that the deviations from RRKM predictions are reaction-specific, but it was not possible to correlate the deviations with specific properties of the reactant. In any event, the average deviation from the RRKM predictions was approximately a factor of 2.4. Our earlier comparisons [6, 7] on fall-off behavior also revealed quite good agreement, even at low temperatures.

Another advantage of using the simpler theory is the ease of formulation for cases where the energized adduct has multiple exit channels [6]. It appears that the QRRK approach represents a suitable compromise for many cases of interest. It allows one to estimate the temperature and pressure dependence of many reactions for inclusion in kinetic models. Prudence dictates that reactions which are critical in such a model be examined more carefully by use of RRKM analysis and/or master equation methods. However, the results described in this paper demonstrate that the QRRK approach allows one to take the first steps toward a more realistic description of the temperature and pressure dependence of addition and recombination reactions.

The QRRK approach will be used in this article to examine several of the methyl radical reactions known to be important in combustion. Given the high concentrations of CH₃ found in combustion environments and the richness of its chemistry, this seems an appropriate choice. The objective is to illustrate how this approach provides a useful method for analysis of these reactions and to suggest specific rate constants for a variety of conditions encountered in combustion.

Methyl radical chemistry can be viewed as a competition between oxidation and recombination. The recombination channel has been studied experimentally [9,10] and by RRKM methods [3]; the simpler QRRK approach has also been applied [6]. The two major oxidation reactions, involving O_2 and O , will be treated here. In addition, the method will be applied to the $CH_3 + OH$ reaction because there is considerable uncertainty regarding the rate and the product distribution for this case. Finally, when the recombination channel is important, the subsequent addition of methyl with acetylene also can become important. This case will also be treated since it illustrates another potential complexity that can be analyzed by the QRRK approach.

Results and Discussion

$CH_3 + O$

This reaction has been extensively studied and has recently been reviewed by Warnatz [11]; there is a consensus that it is very rapid and produces $CH_2O + H$. No pressure effects have been observed, but, in fact, the reaction is a radical combination that forms an excited intermediate CH_3O^* , which dissociates by β -scission. Analysis of this reaction is expected to be quite straightforward. There is only one reaction channel, and complications due to energy transfer should be minimized. Thus, it serves as a relatively easy test of our approach.

It is possible to analyze this reaction within the QRRK chemical activation framework and predict the observed behavior. The energetics are shown in Figure 1, and the equations for the two possible channels (stabilization to CH_3O or reaction to $CH_2O + H$) are [6]:

(A)

$$k_{\text{stab}} = \frac{d[CH_3O]/dt}{[CH_3][O]} = k_1 \sum_{E=-1}^{\infty} \frac{\beta k_s[M]}{\beta k_s[M] + k_{-1}(E) + k_2(E)} \cdot f(E, T)$$

(B)

$$k_{\text{rxn}} = \frac{d[CH_2O]/dt}{[CH_3][O]} = k_1 \sum_{E=-1}^{\infty} \frac{k_2(E)}{\beta k_s[M] + k_{-1}(E) + k_2(E)} \cdot f(E, T)$$

Here k_1 is the rate constant for formation of the complex (the same as the high-pressure rate constant for formation of the stabilized adduct), βk_s is the collision rate constant k_s with efficiency β , and $k_i(E)$ is the unimolecular QRRK dissociation rate constant for path i :

(C)

$$k_i(E) = A_i \frac{n!(n - m_i + s - 1)!}{(n - m_i)!(n + s - 1)!}$$

where A_i is the high-pressure preexponential factor for reaction i , $n = E/h\nu$, i.e., the total number of quanta of energy available,

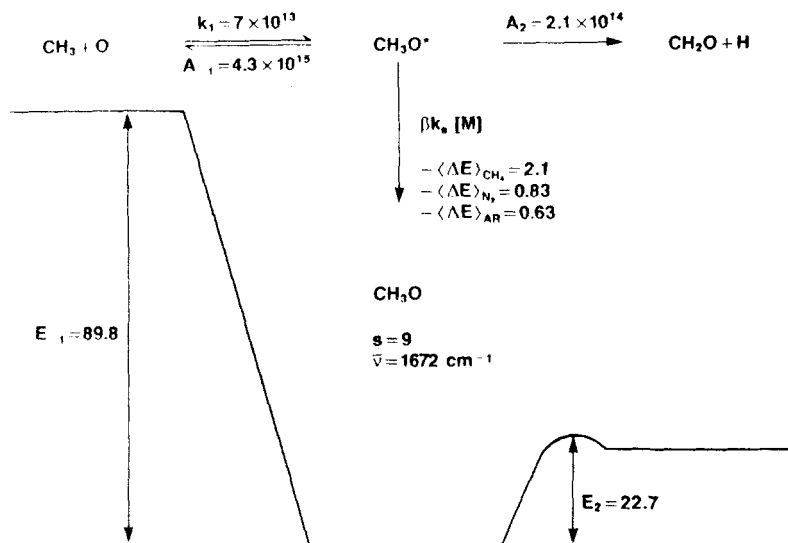


Figure 1. Energy diagram for the $\text{CH}_3 + \text{O}$ reaction; barrier heights and amounts of energy transferred per collision are given in kcal mol^{-1} . The parameters listed were used to obtain the results in Table I and Figure 2.

$m_i = E_i/h\nu$, i.e., the number of quanta corresponding to the energy threshold for reaction i , and s is the total number of vibrational degrees of freedom. The chemical activation distribution function $f(E, T)$ is given by:

$$(D) \quad f(E, T) = \frac{k_{-1}(E)K(E)}{\sum_{E=1}^{\infty} k_{-1}(E)K(E)}$$

$$(E) \quad \text{with } K(E) \approx \alpha^n (1 - \alpha)^s \frac{(n + s - 1)!}{n!(s - 1)!}$$

where $\alpha = \exp(-h\nu/kT)$.

The collisional deactivation efficiency β is calculated as outlined by Troe [12]. This value is based upon a single-channel dissociation mechanism. One might expect more complex behavior in these chemically activated systems with multiple exit channels. A detailed master equation analysis [6] indicated that use of the simplified form is frequently acceptable, particularly for those cases where the various exit channels lie either at similar energies or where there are very large differences in their energies (as is the case for $\text{O} + \text{CH}_3$).

Qualitatively, the low-energy exit channel to $\text{CH}_2\text{O} + \text{H}$ implies that $k_2(E)$ will be very large under all conditions; the pressure would have to be extremely high for the collisional stabilization term $\beta k_s[M]$ to compete. Given the dominance of $k_2(E)$, the rate constant for the reac-

tion channel, k_{rxn} , will approximate k_1 . Thus, the reaction channel would show no dependence on pressure even though it proceeded via a complex. It can be seen from eq. (B) that as long as either of the dissociation rates in the denominator is appreciably larger than the term for collisional stabilization, there will be no pressure dependence. At sufficiently high pressures, one expects an inverse pressure dependence for k_{rxn} [7].

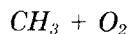
A more quantitative treatment requires assignment of the parameters for explicit evaluation of the apparent rate constants. These parameters are shown in Figure 1. CH_3O has nine oscillators; its geometric mean frequency of 1672 cm^{-1} was obtained from the frequencies listed by Burcat [13]. Calculations were performed assuming a variety of collision partners; the amount of energy transferred per collision by the various partners, $-\langle\Delta E\rangle$, was taken from the compilation by Gardiner and Troe [14]. The collision rate constant, k_s , was based on a Lennard-Jones model with collision diameters and well depths taken from Kee et al. [15]. The rate constant for formation of the complex, $7 \times 10^{13}\text{ cm}^3\text{ mol}^{-1}\text{ s}^{-1}$, was taken to be the average of the measured low-temperature rate constants for the $\text{O} + \text{CH}_3$ reaction [16,17]. As discussed earlier, one expects k_1 to be very similar to k_{rxn} for this reaction. Moreover, this value is quite close to that predicted using Benson's approach [18] for estimating recombination rate constants ($k_{est} = 4.5 \times 10^{13}$). The Arrhenius pre-exponential factor (A -factor) and the barrier for the reverse reaction were obtained by evaluating ΔH_{300}° and ΔS_{300}° (with a standard state of 1 mol/cm^3) for the reaction $\text{CH}_3 + \text{O} = \text{CH}_3\text{O}$ and applying microscopic reversibility.

The A factor and barrier for dissociation to $\text{CH}_2\text{O} + \text{H}$ were obtained by estimating the Arrhenius parameters for the reverse reaction and using the appropriate thermodynamic functions as previously mentioned. It was assumed that the rate constant for $\text{H} + \text{CH}_2\text{O}$ was $4.65 \times 10^{13} \exp(-2.8\text{ kcal}/RT)\text{ cm}^3\text{ mol}^{-1}\text{ s}^{-1}$, a factor of 2 lower (to account for degeneracy) than that for $\text{H} + \text{C}_2\text{H}_4$; the latter was taken from a recent tabulation [19].

Use of the equilibrium constant to obtain the parameters for the dissociation reactions has the advantage of simplicity and reduction in the number of independent parameters. In this article, we chose 300 K as the temperature for evaluation of the equilibrium constant as well for specification of the reverse rate constant. Results from calculations based upon parameters obtained at other temperatures were generally within a factor of 2 of those reported here. Thus, the potential uncertainties introduced by computing Arrhenius-type rate expressions via equilibrium constants that are not exactly described by Arrhenius-type expressions can be quite small.

The results of the calculations are summarized in Table I and compared to the observations in Figure 2. The main uncertainty in the calculations is the experimental uncertainty in k_1 . Within this uncertainty, the QRRK predictions are completely consistent with the data. Of course, this result is hardly surprising, given the very low exit channel for this reaction. Nevertheless, the calculations do present a rationale for the observed but previously unexplained decline in the observed rate constant at high temperatures [20]. This decline is simply a manifestation of complex formation; at high temperatures the energy of CH_3O^* is sufficiently high that dissociation back to reactants (higher A factor and higher barrier) begins to compete with the dissociation channel to products (lower A but also a much lower barrier). Identical results for the reaction channel were obtained with each of the collision partners. Note in Table I that the rate constant for formation of $\text{CH}_2\text{O} + \text{H}$ is virtually independent of pressure between 20–7600 Torr, while the stabilization rate constant scales linearly with pressure over this range. The low exit channel produces low-pressure limit behavior even at pressures as high as 10 atm. The extremely rapid rate for reaction of these species via a complex can be compared to that expected for a possible abstraction reaction to form $\text{CH}_2 + \text{OH}$. This path is 8 kcal mol^{-1} endothermic; even at 2500 K, it would be much slower than the recombination/elimination channel.

Earlier in this article we saw that the presence of a low-energy exit channel resulted in such a short-lived complex that it would not be collisionally stabilized at normal pressures. Secondary decomposition is another manifestation of this low exit channel which can complicate the kinetics. Note that the $\text{CH}_2\text{O} + \text{H}$ products are formed with a large amount of excess energy — 67 kcal mol^{-1} from the difference in barriers plus the internal energy of CH_3O^* . If most of this energy is partitioned into the internal modes of CH_2O , it is possible for the product CH_2O to have sufficient energy to dissociate, giving $\text{H} + \text{CH}_2\text{O}$, $2\text{H} + \text{HCO}$, or $\text{H} + \text{H}_2 + \text{CO}$ as possible products of the $\text{O} + \text{CH}_3$ reaction. The details of the analysis of this secondary decomposition are treated elsewhere [21]. The kinetic implications of such secondary decompositions can be great; they can convert a chain-breaking process into a chain-branching one.



There has been considerable disagreement in the literature regarding the products of this reaction. Several groups [20,22,23] have reported $\text{CH}_3\text{O} + \text{O}$ to be the principal high-temperature channel, while very recent work by Saito et al. [24] suggests that $\text{CH}_2\text{O} + \text{OH}$ is the dominant channel below 1900 K. A potential energy diagram for this

b) $\text{CH}_3 + \text{OH} \rightarrow \text{CH}_2\text{OH} + \text{H}$	Ar	1.94E14 ^b	- 0.45	1625.	1.54E19 ^b	1.74	7572.	5.68E19 ^b	- 1.82	11002.
	N ₂	4.28E14 ^b	- 0.54	1978.	2.64E19 ^b	- 1.80	8068.	1.99E19 ^b	- 1.68	11157.
	CH ₄	3.79E15 ^b	- 0.79	3033.	3.01E19 ^b	- 1.78	9112.	1.73E17 ^b	- 1.07	11051.
	Ar	8.93E11 ^b	- 0.02	13073.	3.87E12 ^b	- 0.19	13741.	5.79E14 ^b	- 0.76	16426.
c) $\text{CH}_3 + \text{OH} \rightarrow \text{CH}_3\text{O} + \text{H}$	N ₂	9.16E11 ^b	- 0.02	13083.	5.74E12 ^b	- 0.23	13931.	9.95E14 ^b	- 0.82	16872.
	CH ₄	9.46E11 ^b	- 0.02	13104.	2.02E13 ^b	- 0.38	14609.	1.08E15 ^b	- 0.80	17820.
	Ar	6.69E13 ^b	- 0.45	1630.	5.39E18 ^b	- 1.74	7581.	1.99E19 ^b	- 1.82	11012.
	N ₂	1.38E14 ^b	- 0.54	1965.	8.90E19 ^b	- 1.80	8067.	6.37E18 ^b	- 1.67	11141.
d) $\text{CH}_3 + \text{OH} \rightarrow \text{CH}_2 + \text{H}_2\text{O}$	CH ₄	1.98E15 ^b	- 0.79	3033.	1.05E19 ^b	- 1.79	9119.	5.71E16 ^b	- 1.07	11043.
	Ar	3.98E10 ^b	- 0.02	8765.	1.60E12 ^b	- 0.45	10449.	6.27E14 ^b	- 1.12	14133.
	N ₂	4.17E10 ^b	- 0.03	8786.	3.19E12 ^b	- 0.53	10810.	7.39E14 ^b	- 1.13	14551.
	CH ₄	5.32E10 ^b	- 0.06	8756.	1.88E13 ^b	- 0.73	10198.	1.76E14 ^b	- 0.93	15240.
e) $\text{CH}_3 + \text{OH} \rightarrow \text{H}_2\text{CO} + \text{H}_2$	Ar	2.43E46 ^b	-10.9	19974.	2.09E41 ^b	- 8.94	20654.	4.29E40 ^c	- 8.41	24274.
	N ₂	1.43E46 ^b	-10.8	20158.	1.61E40 ^b	- 8.58	20331.	1.15E39 ^c	- 7.93	23605.
	CH ₄	8.75E44 ^b	-10.3	20413.	2.68E36 ^b	- 7.40	19020.	5.75E33 ^c	- 6.35	20982.
	Ar	1.00E13	- 0.53	13391.	2.72E18 ^b	- 1.97	20210.	2.48E16 ^b	- 1.32	22134.
4. a) $\text{CH}_3 + \text{HC} \equiv \text{CH} \rightarrow \text{CH}_3\text{CH} = \text{CH}$	N ₂	9.96E12	- 0.53	13542.	2.73E18 ^b	- 1.96	20592.	2.65E15 ^b	- 1.05	21944.
	CH ₄	4.70E12	- 0.42	13831.	4.93E17 ^b	- 1.73	21205.	4.23E19 ^c	- 2.17	27231.
	Ar	1.09E48 ^b	-10.9	28553.	4.32E47 ^c	-10.2	37103.	5.41E38 ^c	- 7.44	36700.
	N ₂	2.06E47 ^b	-10.6	28634.	2.61E46 ^c	- 9.82	36951.	6.29E36 ^c	- 6.87	36120.
b) $\text{CH}_3 + \text{HC} \equiv \text{CH} \rightarrow \text{CH}_3\text{C} \equiv \text{CH}$	CH ₄	4.70E44 ^b	- 9.75	28558.	1.48E42 ^c	- 8.52	36210.	2.03E30 ^c	- 4.99	33558.
	Ar	2.87E21 ^b	- 2.74	24833.	5.72E20 ^b	- 2.36	31471.	1.01E23 ^c	- 2.83	40810.
	N ₂	5.50E21 ^b	- 2.80	25459.	6.74E19 ^b	- 2.08	31591.	2.96E21 ^c	- 2.40	40660.
	CH ₄	7.36E21 ^b	- 2.80	26870.	1.37E25 ^c	- 3.49	38648.	9.56E15 ^c	- 0.86	39240.

^a cal/mol.^b $T = 600\text{--}2500$ K fit.^c $T = 900\text{--}2500$ K fit.^d Experimental average [23], $T = 1150\text{--}2300$ K.

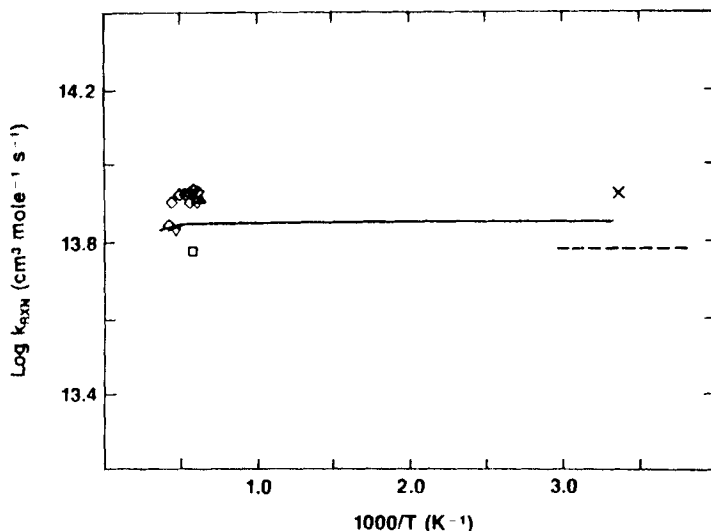
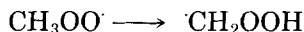


Figure 2. Comparison of the bimolecular QRRK predictions and observations for $\text{CH}_3 + \text{O} \rightarrow \text{CH}_2\text{O} + \text{H}$. Solid line is the QRRK prediction. X = Washida; --- = Washida and Bayes; Δ = Biordi; ∇ = Olson and Gardiner; $\diamond$ = Bhaskaran et al.; $\square$ = Tsuboi. (References to observations are given by Warnatz [11]).

reaction is shown in Figure 3. Hsu et al. [23] performed a detailed RRKM analysis and rationalized their observation of $\text{CH}_3\text{O} + \text{O}$ by assuming a barrier of 60 kcal mol^{-1} for the intramolecular transfer:



However, this analysis is incompatible with the observations of Saito et al. which require a lower barrier. Saito et al. calculated an *ab initio* barrier of 49 kcal mol^{-1} , but even this value was too high to rationalize their results. By analogy to other 1,3 hydrogen shifts [25], one can estimate a barrier of approximately 37 kcal mol^{-1} (26 kcal for ring strain, 6 kcal for the H-abstraction, and 5 kcal for the reaction endothermicity). We felt it would be worthwhile to include a barrier of this magnitude in our calculations to better assess the possibility of the hydrogen shift. Inclusion of the isomerization channel is straightforward; details are given in ref. [6].

Note the energetics of the $\text{CH}_3 + \text{O}_2$ case are markedly different from $\text{CH}_3 + \text{O}$. Here the higher-energy exit barriers require that the complex have appreciable thermal energy in addition to the energy arising from bond formation before these channels become significant. The absence of a low-energy exit channel with a low barrier suggests that significant stabilization of the complex could occur.

The parameters required for application of the QRRK method were obtained as described earlier. We adopt the value recently measured by

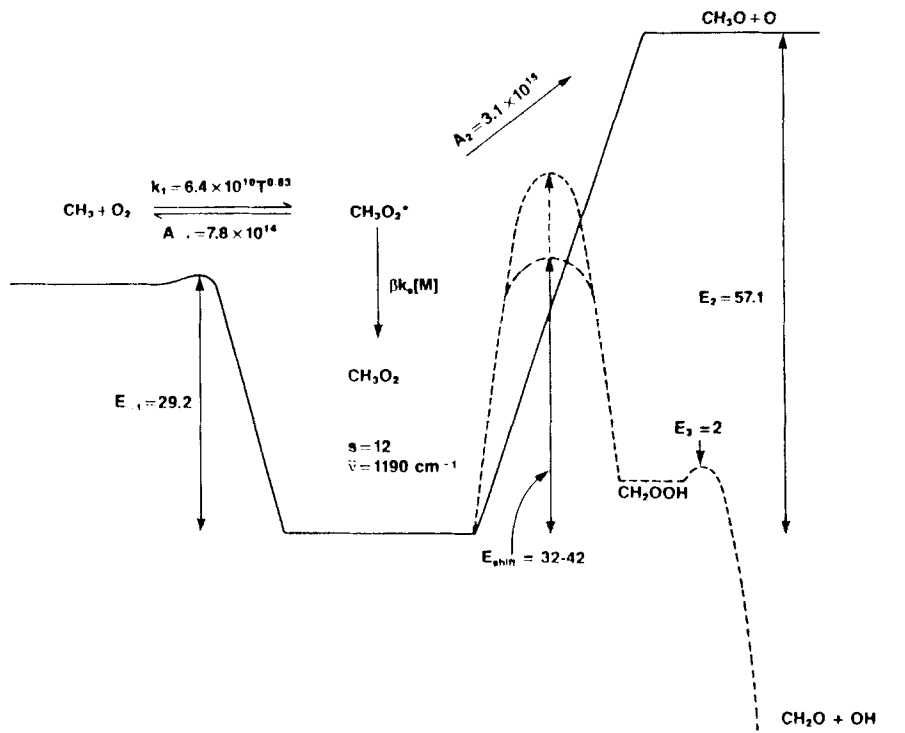


Figure 3. Energy diagram for the $\text{CH}_3 + \text{O}_2$ reaction; barrier heights are given in kcal mol^{-1} . The parameters listed were used to obtain the results in Table I and Figures 4, 5, and 6.

Cobos et al. [26] for k_1 . The A factor and barrier for dissociation of the complex back to reactants were obtained via ΔH_{300}° and ΔS_{300}° . (The expression for k_1 was transformed into Arrhenius form to obtain these values.) The parameters for dissociation to $\text{CH}_3\text{O} + \text{O}$ were obtained using an estimate of $5 \times 10^{13} \text{ cm}^3 \text{ mol}^{-1} \text{ s}^{-1}$ for the high-pressure recombination rate constant for the reverse reaction. This value is based upon the assumption that the steric properties of CH_3O are similar to C_2H_5 ; using Benson's [18] estimates for differences in the steric factor between CH_3 and C_2H_5 suggests this rate constant is 25% lower than the $\text{O} + \text{CH}_3$ value. The A factor for intramolecular H-transfer ($6 \times 10^{12} \text{ s}^{-1}$) was similar to that suggested by Benson [25]. The A factor for the scission of CH_2OOH ($5.5 \times 10^{13} \text{ s}^{-1}$) was based upon the assumption that the rate constant for OH addition to CH_2O was similar to that recently measured [27] for $\text{OH} + \text{CH}_3\text{CHO}$. A low barrier of 2 kcal mol^{-1} was assumed, consistent with the low intrinsic activation energies expected for OH addition. The geometric-mean frequency was calculated from Burcat [13]. The same collision partners were used as for the previous case.

Predictions of this model are compared to some recent observations [26,28] of room temperature fall-off behavior in N_2 for the stabilization channel in Figure 4. The fit is quite good above 0.1 atm, but it falls below the observations at lower pressures, resulting in an underestimate of the apparent rate constant at the lowest pressures by about a factor of 3. As expected, the stabilization channel is much more important for this reaction where lack of low-energy exit channel results in sufficiently long lifetimes of the energized adduct that stabilizing collisions occur. These results are included in Table I.

The predicted high-temperature rate constant for the $O + CH_3O$ channel is compared to experiment in Figure 5. The agreement is very good. The predicted rate constants for both the stabilization channel and the $O + CH_3O$ channel are very insensitive to the presence or absence of the $CH_2O + OH$ channel. Furthermore, the rate constants for $CH_3O + O$ are quite insensitive to collision partner or pressure in the 20–7600 Torr range.

The specific prediction for the $CH_2O + OH$ channel depends strongly on the isomerization barrier, as shown in Figure 6. Given the uncertainties in estimating this barrier, predicted values are shown using barriers of 32, 37, and 42 kcal mol $^{-1}$. (These predictions are quite insensitive to collision partner or pressure between 20–7600 Torr.) It is readily apparent that the calculations do not permit a choice between

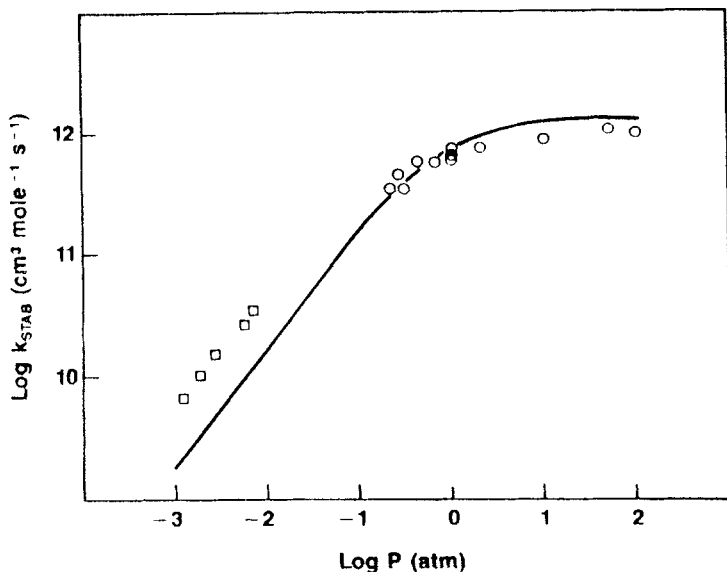


Figure 4. Comparison of predicted QRRK rate constants (solid line) versus observed ($\circ$ = Cobos et al. [26]; $\square$ = Selzer and Bayes [28]) for $CH_3 + O_2 \rightarrow CH_3O_2$ at room temperature in N_2 . Predictions based upon parameters given in Figure 3.

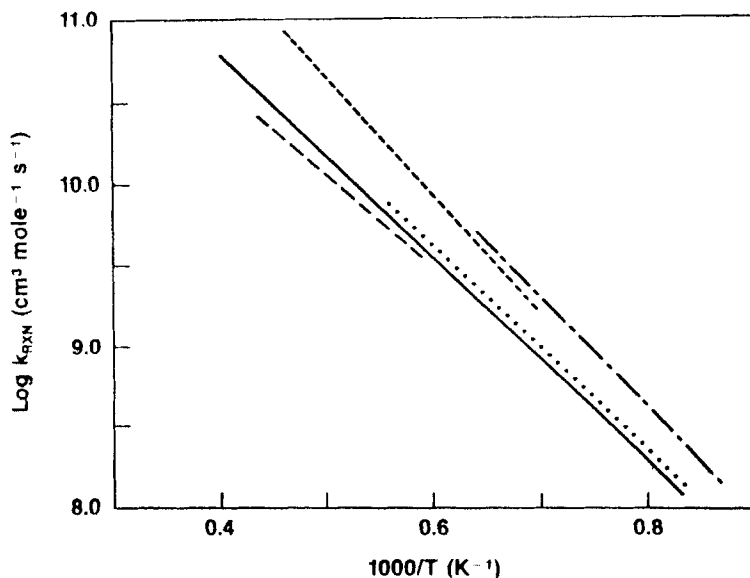


Figure 5. Comparison of predicted QRRK rate constants (solid line) versus observed — — — = Hsu et al. [23]; — — — = Bhaskaran et al. [20]; ····· = Brabbs and Brokaw [22]; — — — — = Saito et al. [24] for $\text{CH}_3 + \text{O}_2 = \text{CH}_3\text{O} + \text{O}$. Calculated values shown are for 1 atm with N_2 as collision partner.

the various sets of experimental results. The upper limits reported by Bhaskaran et al. [20] or Baldwin and Golden [29] suggest a barrier above 40 kcal mol⁻¹, while the higher values reported by Saito et al. [24] are more consistent with a lower barrier, nearer 32 kcal mol⁻¹. This unfortunate ambiguity in the predictions is not traceable to the QRRK approach, but rather to the uncertainty in the isomerization barrier.

The addition of CH_3 to O_2 illustrates many of the features expected in alkyl additions. Of particular interest with the larger alkyl radicals is the possibility of the intramolecular H shift to yield an olefin and HO_2 upon dissociation. The barrier to such a shift would be expected to be smaller than for the methyl case since there would be less strain. However, one also expects a lower A factor since there would be a larger loss of entropy in the transition state. These values are readily estimated for a particular case [25], and one can then predict the effect of both temperature and pressure on the ratio of stabilization versus olefin production. Such predictions can be calibrated by comparison to recent experimental measurements of these reactions by Gutman and coworkers [30]. Similarly, one can estimate the effect of addition of unsaturated radicals to oxygen where cyclization of the initial adduct can lead to breaking of the multiple bond. Again one has the luxury of comparison to recent data [31] to validate the kinetic parameters for the cyclization reaction.

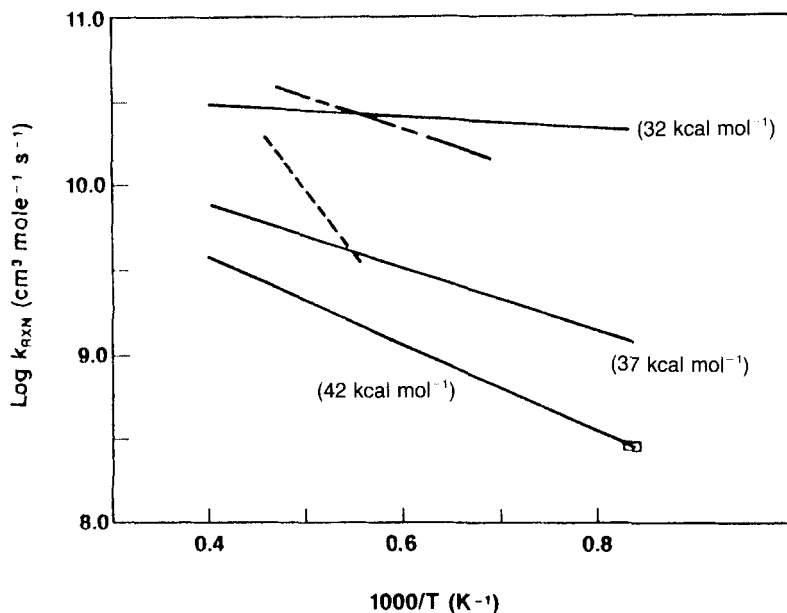


Figure 6. Comparison of predicted QRRK rate constants versus observed — · — · = Saito et al., [24] — — — = Bhaskaran et al. [20] (upper limit); □ = Baldwin and Golden [29] (upper limit) for $\text{CH}_3 + \text{O}_2 = \text{CH}_2\text{O} + \text{H}$. The three solid lines represent the predicted values with three different barriers for the isomerization reaction $\text{CH}_3\text{OO} \rightarrow \text{CH}_2\text{OOH}$. Calculated values shown are for 1 atm with N_2 as the collision partner.

$\text{CH}_3 + \text{OH}$

There is sufficient uncertainty regarding this reaction that Warnatz [11] refused to recommend a value. However, given its expected similarity to its isoelectronic analog $\text{CH}_3 + \text{CH}_3$, we calculated apparent rate constants for its possible channels. Chemically activated CH_3OH has the opportunity to dissociate in a wide variety of ways. There are two ways to lose hydrogen atoms and there are two possible concerted reactions—one involving H_2 elimination and one involving H_2O elimination. An energy level diagram for these various channels is shown in Figure 7. The value of k_1 is an estimate based upon the assumption that this rate constant should be similar to $\text{CH}_3 + \text{CH}_3$. Parameters for the hydroxymethylene (CH_2OH) + H and the methoxy (CH_3O) + H channels are based upon an estimated value of $6.4 \times 10^{13} \text{ cm}^3 \text{ mol}^{-1} \text{ s}^{-1}$ for the reverse recombinations. The 102 kcal mol⁻¹ barrier for the H_2 elimination is based upon a calculated barrier of 80 kcal mol⁻¹ for the reverse (addition) reaction [32]. The corresponding A factor assumes a tight transition state. The parameters for production of singlet methylene and water are based upon measurements of the reverse reaction at

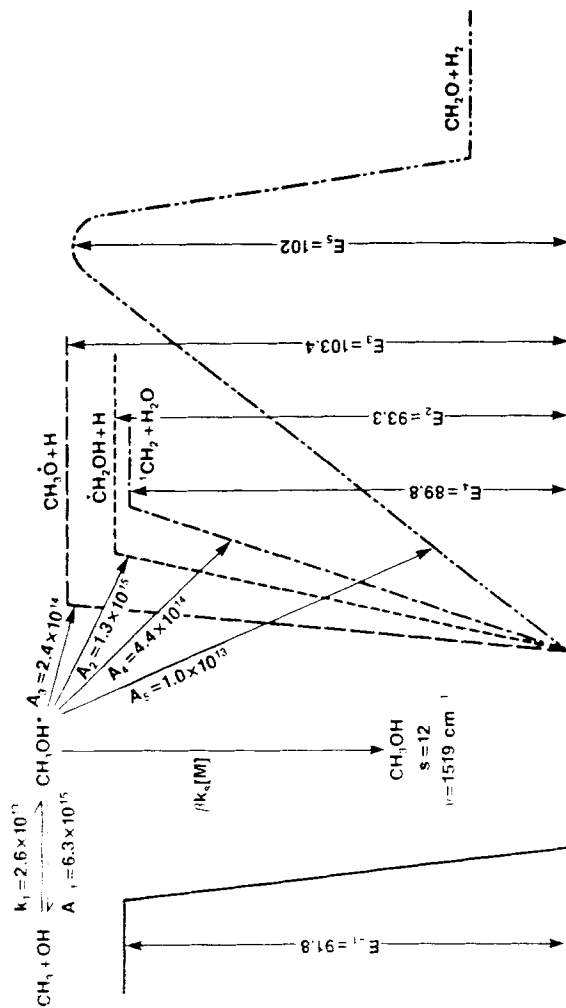


Figure 7. Energy diagram for the $\text{CH}_3 + \text{OH}$ reaction; barrier heights are given in kcal/mol. The parameters listed were used to obtain the results in Table I and Figure 8.

1 atm and 298 K [33]. The geometric mean frequency was calculated from those listed by Burcat [13].

Calculations for 1 atm Ar (Fig. 8) indicate that collisional stabilization for CH_3OH dominates up to 1400 K, but that chemically activated decomposition to $\text{H} + \text{CH}_2\text{OH}$ is faster at higher temperatures. For completeness, rate constants for the two likely abstraction channels for $\text{CH}_3 + \text{OH}$ are also included in Figure 8. The estimate for formation of triplet methylene and water is based upon analogy to triplet methylene abstraction of alkyl hydrogens by Dobe et al. [34] using their general A factor and Evans-Polanyi plots. The reaction to form $\text{CH}_4 + \text{O}$ is based upon measurements of the reverse rate constant [11] and microscopic reversibility. Note that at temperatures near 2000 K, the products of the reaction reflect a mixture of pressure-dependent addition (CH_3OH , $\text{CH}_2\text{OH} + \text{H}$, and $^1\text{CH}_2 + \text{H}_2\text{O}$) and

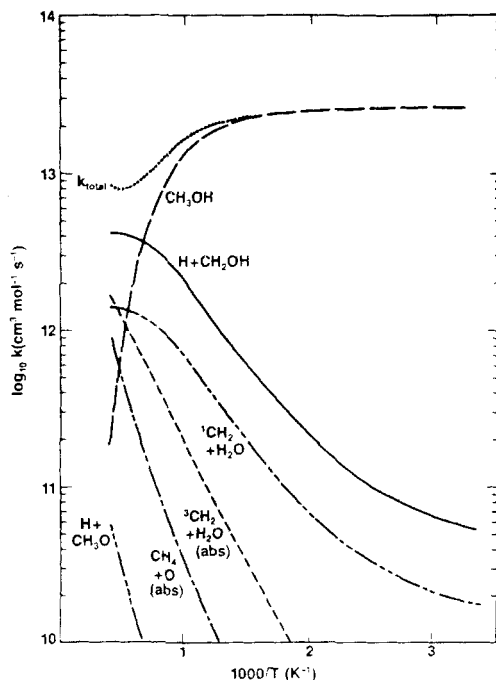


Figure 8. Predicted values for the rate constant for the various channels of the $\text{CH}_3 + \text{OH}$ reaction. Results are given for a pressure of 760 Torr with Ar as the collision partner. = k_{tot} , sum of the rate constants for all the channels;

- = $k(\text{CH}_3 + \text{OH} \rightarrow \text{CH}_3\text{OH})$;
- = $k(\text{CH}_3 + \text{OH} \rightarrow \text{CH}_2\text{OH} + \text{H})$;
- - - = $k(\text{CH}_3 + \text{OH} \rightarrow ^1\text{CH}_2 + \text{H}_2\text{O})$;
- - - = $k(\text{CH}_3 + \text{OH} \rightarrow ^3\text{CH}_2 + \text{H}_2\text{O})$ (abstraction);
- - - = $k(\text{CH}_3 + \text{OH} \rightarrow \text{O} + \text{CH}_4)$ (abstraction);
- - - = $k(\text{CH}_3 + \text{OH} \rightarrow \text{H} + \text{CH}_3\text{O})$.

pressure-independent abstractions ($^3\text{CH}_2 + \text{H}_2\text{O}$ and $\text{CH}_4 + \text{O}$). When a low-pressure flame is considered, $\text{CH}_2\text{OH} + \text{H}$ dominates at even lower temperatures. At no temperature or pressure are the channels to $\text{CH}_2\text{O} + \text{H}_2$ or $\text{CH}_3\text{O} + \text{H}$ important, principally because of their higher barriers.

An indirect test of the $\text{CH}_3 + \text{OH}$ predictions is to compare a unimolecular QRRK description to the data of Spindler and Wagner [35]. Their rate constants for dissociation of CH_3OH to $\text{CH}_3 + \text{OH}$ were measured in the fall-off region from 0.13 to 6.5 atm. over the temperature range 1600–2000 K. QRRK calculations, using the unadjusted parameters of Figure 7, predicted rate constants in good agreement with the shape and magnitude of these data, being higher only by factors of 1.6 to 2.8.

An added complexity here again is the possibility of secondary decomposition. Unlike the $\text{O} + \text{CH}_3$ case considered earlier, there is no low exit channel which would leave the products with appreciable excess energy. However, the products in the present case have much lower barriers for decomposition than the CH_2O formed earlier. For example, the β -scission reaction of CH_2OH to give $\text{CH}_2\text{O} + \text{H}$ only requires 31 kcal mol⁻¹. At high temperatures, the still-excited CH_2OH radical could contain sufficient energy to undergo this reaction rather than being collisionally stabilized. Even if the radical is stabilized, the 31 kcal mol⁻¹ bond in the thermalized adduct will decompose very rapidly. The result in either case is conversion of methyl and hydroxyl radicals to two hydrogen atoms. The branching ratio between stabilization and reaction of the initial adduct would be unaffected by any secondary decomposition. The only effect of secondary decomposition would be a change in the product distribution within a particular reaction channel.

$\text{CH}_3 + \text{C}_2\text{H}_2$

This reaction can be quite important in high-temperature environments since C_2H_2 is a dominant hydrocarbon species at high temperatures. Moreover, as shown later, this reaction provides a facile route for production of the resonantly stabilized allyl radical. It has been proposed that this long-lived radical could then play a significant role in molecular-weight growth via addition reactions to unsaturated hydrocarbons [36].

The calculations mentioned previously have illustrated that new dissociation channels can open up if there is sufficient energy in the complex. Similarly, it was shown earlier [6] that there could be enough energy to allow the complex to isomerize before stabilization. The specific case of isomerization in this reaction via a 1,3-hydrogen shift was considered using generic estimates for the various rate constants.

There it was shown that such a shift might be possible if the barrier were as low as 20 kcal mol⁻¹.

A more rigorous assessment is made here, including updated estimates of the *A* factor for the 1,3-hydrogen shift, as well as use of a more realistic barrier of 36 kcal mol⁻¹. The potential energy diagram, based on a bond dissociation energy of 110 kcal mol⁻¹ [37] for C₂H₃—H, is shown in Figure 9, along with the necessary parameters. (Use of a lower value of 103 kcal mol⁻¹ as suggested by a recent kinetic study [38] had relatively little effect.) The rate constant for complex formation is that reported by Holt and Kerr [39]. The parameters for the 1,3-hydrogen shift are based on thermochemical estimates. The parameters for production of CH₃C≡CH + H and H₂C=C=CH₂ + H are based upon the appropriate equilibrium constants and measured values for the reverse reactions [$k(\text{H} + \text{CH}_3\text{C}\equiv\text{CH}) = 5.8 \times 10^{12} \exp(-3100/RT)$, $k(\text{H} + \text{CH}_2=\text{C}=\text{CH}_2) = 4.0 \times 10^{12} \exp(-2700/RT)$]¹⁹. The geometric mean frequency was computed using Burcat's [13] values. The predicted rate constants for the various channels at 1 atm with CH₄ as the collision partner are shown in Figure 10. Here the two stabilization channels dominate; note that there is a relatively high probability of directly forming allyl under these conditions, particularly above 1500 K. These results are quite insensitive to the kinetic parameters for the 1,3-H shift. For example, changing the barrier height by 5 kcal mol⁻¹ changes the apparent rate constant for allyl production by about a factor of 2 at 1500 K; other channels show smaller changes. Similarly, increasing the *A* factor for the shift by a factor of 2 results in an increase of only half that for allyl production at 1500 K.

These calculations might provide a plausible explanation for the 1,3-hydrogen shifts that have frequently been invoked to attempt to explain various kinetic observations. The key factor is recognition of the fact that the complex might have sufficient energy, prior to stabilization, to surmount the barrier for isomerization, just by nature of the way in which it was formed. This is just another example of the importance of chemical activation in high-temperature kinetics. The energy deposited in an energized complex, both from the bond that is formed (chemical activation) and the thermal energy of the components, allows these adducts to react in ways that would not be possible if they were thermalized.

Conclusions

The examples considered earlier illustrate many of the complexities of methyl radical reactions. They also illustrate a method for dealing with this complexity. It seems clear that one can get a reasonable feel for the chemistry by considering the reaction energetics. The param-

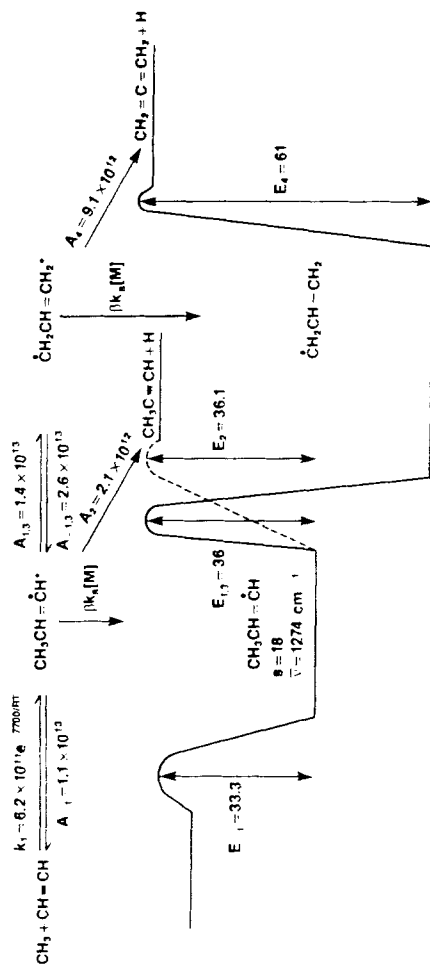


Figure 9. Energy diagram for the $\text{CH}_3 + \text{C}_2\text{H}_2$ reaction; barrier heights are given in kcal mol^{-1} . The parameters listed were used to obtain the results in Table I and Figure 10.

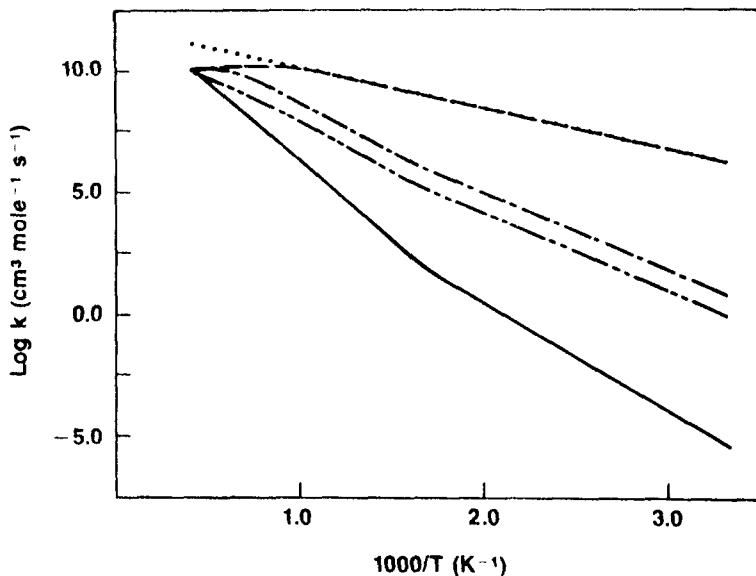


Figure 10. Predicted values for the rate constant for the various channels of the $\text{CH}_3 + \text{C}_2\text{H}_2$ reaction. Results are given for a pressure of 760 Torr with CH_4 as the collision partner.

- = high-pressure rate constant for production of $\text{CH}_3\text{CH} = \dot{\text{C}}\text{H}$.
- = $k(\text{CH}_3 + \text{C}_2\text{H}_2 \rightarrow \text{CH}_3\text{CH} = \dot{\text{C}}\text{H})$;
- · - · - = $k(\text{CH}_3 + \text{C}_2\text{H}_2 \rightarrow \dot{\text{C}}\text{H}_2\text{CH} = \text{CH}_2)$;
- - - - = $k(\text{CH}_3 + \text{C}_2\text{H}_2 \rightarrow \text{CH}_3\dot{\text{C}} \equiv \text{CH} + \text{H})$;
- = $k(\text{CH}_3 + \text{C}_2\text{H}_2 \rightarrow \text{CH}_2 = \text{C} = \text{CH}_2 + \text{H})$

ters needed to put this "feel" on a semi-quantitative basis are readily obtained; it is then quite easy to predict the temperature and pressure dependence for the various product channels. Certainly these calculations are no substitute for experiments. Nevertheless, the agreement with experiment is quite encouraging given the ease of generating the estimates and the fact that there were no parameter adjustments. The kinetic parameters were obtained via equilibrium relationships from measured or estimated values of the reverse rates. The collisional stabilization rates were based upon tabulated Lennard-Jones parameters and tabulated values of the energy transferred per collision. The non-kinetic properties of the energized complex were based on tabulated frequencies and the total number of vibrations.

The possibility of dissociation of an energized complex via either low- or high-energy exit channels requires that the usual thinking about reactions which form adducts be modified. In particular, it is not correct to assume that measurement of a pressure-independent rate constant means that a reaction is at the high-pressure limit. The key is to distinguish the pressure dependence of the product channels. High-pressure limiting behavior is achieved only when the stabilization

channel is pressure-independent. It has been shown earlier in this article that pressure independence of the reaction channel is evidence that the system is actually at the low-pressure limit for stabilization.

The reactions considered in this article are illustrative of many such reactions that could be of importance in combustion chemistry. All recombination, addition, and insertion reactions proceed via formation of a complex. The high temperatures result in a large thermal energy contribution added on top of the energy resulting from bond formation; therefore, a disconcertingly large number of reaction channels become accessible. This behavior introduces an unavoidable complication to the combustion modeler. The approach described in this article presents a relatively easy approach to estimate these various paths. It tilts toward simplicity as much as possible, and cannot be expected to do as well as more sophisticated treatments. Nevertheless, comparisons in this paper and elsewhere [7] suggest the predictions are sufficiently accurate to warrant inclusion in kinetic models when no experimental data are available. It allows the modeler to address a large number of such reactions, obtain estimates of the rate constants for the significant channels, and thus solve the kinetic model. This solution can then be subjected to sensitivity analysis; reactions of crucial importance can then be examined more carefully.

Acknowledgments

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